

Micosten Cream 2%

Description: A white, smooth homogenous cream.

Composition: Each gram contains Miconazole Nitrate (Micronised) 20mg.

Pharmacodynamic:

Miconazole Nitrate is a synthetic antifungal agent which inhibits the growth of the common dermatophytes. *Trichophyton rubrum*, *Trichophyton mentagrophytes* and *Epidermophyton floccosum*, the yeast-like fungus, *Candida albicans* and the organism responsible for tinea versicolor. (*Malassezia furfur*).

Fungistatic: may be fungicidal, depending on concentration. Inhibits biosynthesis of ergosterol or other sterols, damaging the fungal cell wall membrane and altering its permeability. As a result, loss of essential intracellular elements may occur. Also inhibits biosynthesis of triglycerides and phospholipids by fungi; in addition, inhibits oxidative and peroxidative enzyme activity, resulting in intracellular buildup of toxic concentrations of hydrogen peroxide, which may contribute to deterioration of subcellular organelles and cellular necrosis. In *Candida albicans*, inhibits transformation of blastospores into invasive mycelial form. Systemic absorption following topical application to normal skin has not been reported.

Pharmacokinetics:

Absorption

Miconazole remains in the skin after topical application for up to 4 days. Systemic absorption of miconazole is limited, with a bioavailability of less than 1% following topical application of miconazole. Plasma concentrations of miconazole and/or its metabolites were measurable 24 and 48 hours after application.

Systemic absorption has also been demonstrated after repeated application of miconazole to infants with diaper dermatitis. Plasma levels of miconazole were undetectable or low in all infants.

Distribution

Absorbed miconazole is bound to plasma proteins (88.2%) and red blood cells (10.6%).

Metabolism and Excretion

The small amount of miconazole that is absorbed is eliminated predominantly in feces as both unchanged drug and metabolites over a four-day post-administration period. Smaller amounts of unchanged drug and metabolites also appear in urine.

Indication:

For topical application in the treatment of tinea pedis (athlete's foot), tinea cruris, and tinea corporis caused by *Trichophyton rubrum*, *Trichophyton mentagrophytes*, and *Epidermophyton floccosum*. Also indicated in the treatment of cutaneous candidiasis

(moniliasis) caused by *Candida albicans*. Topical miconazole nitrate is also indicated in the treatment of tinea versicolor caused by *Malassezia furfur*.

Contraindication:

Contraindicated in patients who are sensitive to topical miconazole.

Side effects:

Irritation, burning, maceration, allergic contact dermatitis and other sign of skin irritation not present before therapy are some of the isolated reaction reported.

Warnings and Precautions:

When used in the treatment of candidiasis, occlusive dressings should be avoided since they provide conditions that favor growth of yeast and release of its irritating endotoxin. Use of topical antifungal may lead to skin sensitization, resulting in hypersensitivity reactions with subsequent topical or systemic use of the medication. If the cream is used on intertriginous areas, it should be used sparingly and massaged in well to avoid maceration effects. To reduce the possibility of recurrence, comply with full course of therapy. Consult the doctor or pharmacist if there is no improvement after four weeks of treatment. Avoid contact with the eyes.

In patients on warfarin, caution should be exercised and the anticoagulant effect should be monitored (see Interactions)

Pregnancy and lactation:

Use with caution in Pregnancy and lactation

Drug interaction:

Miconazole administered systemically is known to inhibit CYP2C9 enzyme system. Due to the limited systemic availability after topical application, clinically relevant interactions occur very rarely. In patients on warfarin which is subjected to metabolism by CYP2C9, caution should be exercised and the anticoagulant effect should be monitored (see Warnings and Precautions).

Effects on ability to drive and use machines

Not applicable.

Route of administration:

Topical administration

Dosage & administration:

Apply to cover the affected area twice daily in patients with tinea versicolor. Twice daily (morning and evening) in patients with tinea pedis, tinea cruris, tinea corporis and

cutaneous candidiasis. When all lesions have disappeared (usually after 2-5 weeks), prolong treatment for some 10 days to prevent relapse.

Overdosage & treatment:

Acute overdosage with topical miconazole is unlikely. If irritation or sensitivity reactions occur, use of the medication should be discontinued. In the event of accidental ingestion, an appropriate method of gastric emptying may be used if considered desirable.

Presentation: Aluminium tubes of 5, 10, 15, and 50gm.

Storage Conditions:

Store in a cool dry place below 30°C. Keep the containers tightly closed. Protect from light and freezing. For external use only. Keep medicine out of reach of children.

Manufactured by:

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